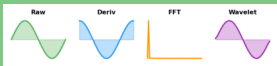


Input Data

- Time series, spatial, or multi-channel data
- Shape: (n_samples, n_channels, n_timepoints)
- Supports variable-length sequences



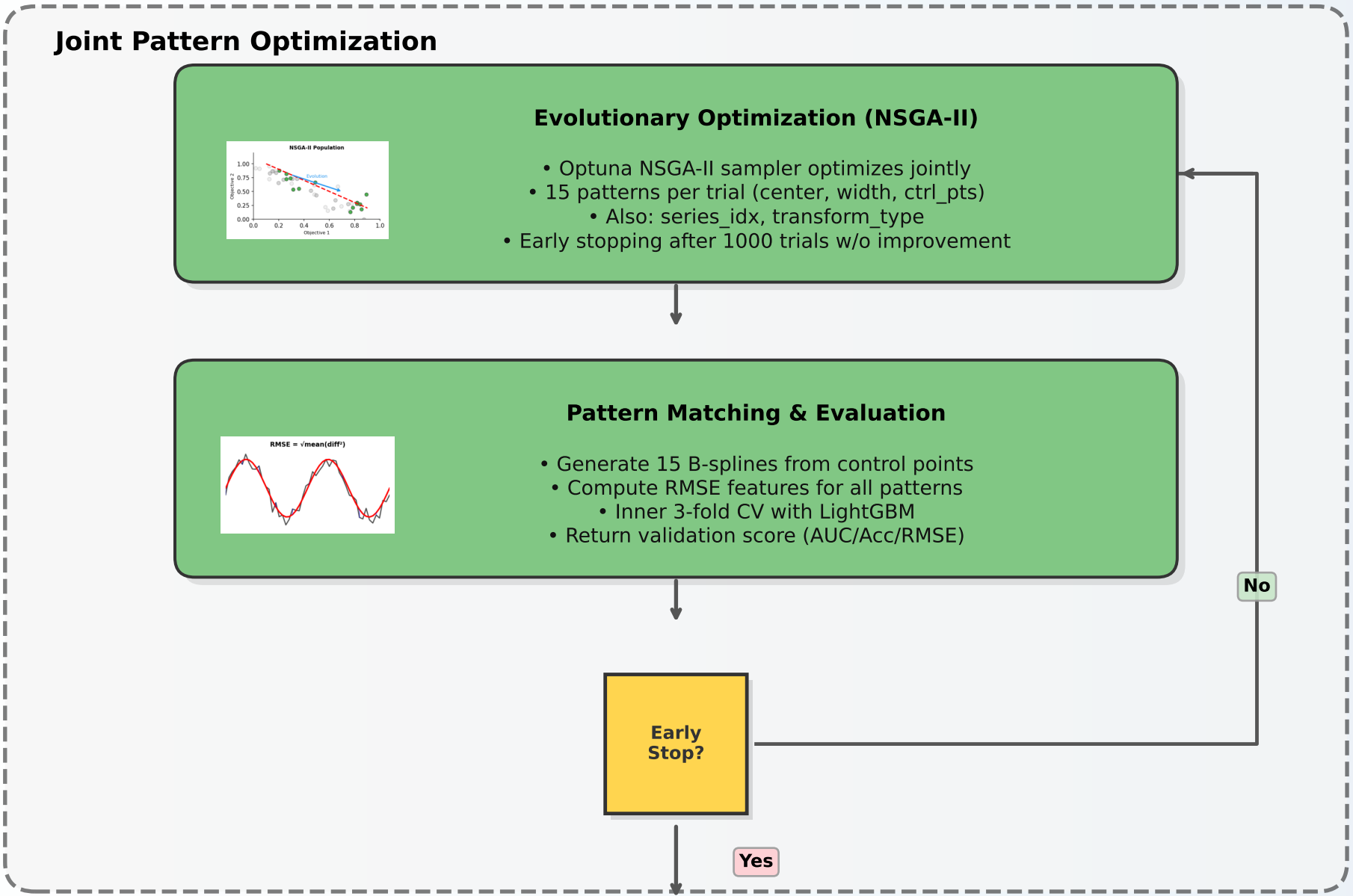
Precompute Transformations

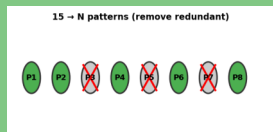
- 21 transform types applied to all channels
- Includes: Raw, Derivative, FFT Power, DCT
 - Wavelets: db4, sym4, coif1, haar
- Also: Autocorr, Reciprocal, Cumsum, Diff...



Transform Selection

- Compute aggregate stats per transform
- Train LightGBM model for each transform
 - 3-fold CV for stable performance
- Select Top-5 transforms by CV score





Backward Elimination

- Start with 15 optimized patterns
- Iteratively remove least useful pattern
- Stop when removal hurts performance
 - Yields compact final pattern set



Final Output

- List of extracted patterns with metadata
 - Trained LightGBM model
 - Train/test feature matrices